



Powering future urban transport: a renewable-based electrification model for public transit—A Mediterranean case study

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ABSTRACT

Transportation is a major contributor to Greenhouse Gas emissions and climate change. To achieve net-zero targets, global policies increasingly promote transport electrification combined with renewable energy integration. This study proposes a scalable methodology for decarbonizing Local Public Transport (LPT) systems through the electrification of services supported by photovoltaic (PV)-based microgrids and storage systems. The methodology begins with the definition of routes, stops, and vehicle schedules, considering commercial speeds and consumption patterns to determine fleet size and energy demand. Based on this demand, a PV-powered microgrid, optimally sized using HOMER software and supplemented with storage, is designed to ensure a continuous supply, including nighttime charging. A case study in Alexandria, Egypt, characterized by traffic congestion and poor air quality, demonstrates the approach. Two new depots with PV capacities of 12.143 MW and 8.873 MW were modeled, capable of powering the entire service while generating profitable excess electricity. The system is projected to save 17,544 tons of CO₂ annually, corresponding to \$1.4 million in avoided emission costs. Sensitivity analysis varying speeds, frequencies, and irradiance confirms the system's robustness across diverse scenarios. Simulations reduce waiting times to 10 min, enhancing service quality. The methodology offers a practical and replicable strategy for creating 100 % green, well-to-wheel zero-emission transport systems. Its scalability allows adaptation to different urban contexts, while excess electricity can further enhance economic sustainability. Although the absence of passenger demand data limits precision, the approach provides a robust framework to guide urban mobility electrification in support of national net-zero objectives.

1. Introduction

Electricity generation and transportation stand out as prominent sectors contributing to the escalation of Greenhouse Gases (GHGs) in the atmosphere. Carbon dioxide (CO₂) emerges as one of the primary gases with the most substantial GHG impact on the environment. The electricity generation sector is responsible globally for the emission of 36.6 GtonsCO₂, while transportation, second in this ranking, emits 7.7 GtonsCO₂ [1]. Nowadays, in order to deal with climate change issues, various authorities are pushing towards the decarbonization of various sectors, including them [2]. While the decarbonization of the electricity generation sector is underway in many countries, the decarbonization of transportation is still behind [3].

Nomenclature

GHG	Green House Gase	SoH	State of Health
LCA	Life Cycle Assessment	i_{BSS}	BSS current
EV	Electric Vehicle	C	Useable BSS capacity
PTO	Public Transport Operator	CC	Charger Controller
LPT	Local Public Transport	MPPT	Maximum Power Point Tracking
EVSE	Electric Vehicle System Equipment	P_{load}	Power required by the load
RES	Renewable Energy Source	E_{load}	Energy required by the load
AQI	Air Quality Index	P_{grid}	Power of the grid
V2G	Vehicle to Grid	P_{PV}	Power generated by PV
ICE	Internal Combustion Engine	$P_{storage}$	Power of the storage system
L	Length of the route	P_{loss}	Power losses
CC	Charger Controller	P_{peak}	Peak power of the load
MPPT	Maximum Power Point Tracking	ϵ	Inverter efficiency

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GHI	Global Horizontal Irradiance	Li-ion	Lithium Ion
N_{maxst}	Maximum number of stations	V_n	Nominal battery voltage
f_l	Length factor	Cap_n	Nominal battery capacity
f_p	Percentage factor	ϵ_{rt}	Battery roundtrip efficiency
BRT	Bus Rapid Transit	NPC	Net Present Cost
LRV	Light Rail Vehicle	c_{in}	Investment cost
Max_{BRT}	Number of buses to perform the daily BRT service	c_{op}	Operating cost
N	Number of buses to meet the service demand	c_{fuel}	Fuel cost
P_{PV}	PV power production	V_{pres}	Present value
P_{rated}	Rated power of 1 PV plate	V_{Rev}	Revenues value
$Prod_{avg}$	Global solar irradiance	CoE	Cost of Energy
I_g	Global horizontal irradiance	$C_{ann_{tot}}$	Total annualized cost
P_{PV}^n	Total rated power	E_{served}	Energy served
η_{pan}	PV panel efficiency	f_{CO_2}	Average grid emission factor
η_{inv}	Inverter efficiency	CSV	CO ₂ Saving Value
ω_{deg}	Degradation coefficient	P_{CO_2}	CO ₂ emission cost
I_{STD}	Irradiance in standard condition	P_{depot_i}	Power produced in the i-depot
BSS	Battery Storage System	A_{depot_i}	i-Depot area
DoD	Depth of Discharge	P_{tot}	Total power of the equivalent depot
SoC	State of Charge		

The need for a significant revolution in transportation was recognized by public authorities, who are not only promoting electric mobility but also advocating for enhancements in public transportation services. The proposed shift aims to.

- Decarbonize and renew outdated and polluting vehicle fleets;
- Increase the quality of the service to increase users;
- Improve the quality of life in increasingly large urban areas.

By encouraging the use of public transport would make it possible to reduce traffic congestion in urban areas, improving the quality of life of citizens, reducing car accidents and improving air quality. Following this logic, many studies on the positive effects of decarbonization of the sector have been carried out and many urban areas are equipping themselves with electrified vehicle fleets to comply with the new policies [4]. However, it is worth specifying that the carbon footprint of this new type of vehicle should not be overlooked. Life Cycle Assessment (LCA) studies ensured that the complete carbon footprint of Electric Vehicles (EVs) is clear, while highlighting the positivity of the shift to the electric motor. One of the main aspects linked to the environmental sustainability of electric vehicles is linked to the production of electricity they use. This concern is linked to the fact that electricity generation is the most responsible sector for CO₂ emissions [5].

1.1. Literature review: decarbonization strategies

Growing awareness of climate change has driven governments to adopt long-term decarbonization strategies, focusing on restructuring energy systems and electrifying end-use sectors, particularly transport [6]. Multi-model analyses show that current Paris Agreement actions still leave an emissions gap of 22–28 GtCO₂eq by 2030, underscoring the need for stronger policies such as carbon taxes and renewable incentives [7]. European pathways highlight net-zero by 2050, enabled by EV adoption, energy efficiency, and renewable integration [8], while urban-scale studies emphasize air quality and social co-benefits of transport decarbonization [9]. Case studies in Mexico [10], Costa Rica [11], and Ecuador [12] show national strategies centered on renewable expansion and EV growth, while oil-producing Oman [13] and the UK [14] explore electrification challenges and opportunities such as V2G. Similar efforts in China [15] and emerging work in developing countries [16] demonstrate that both industrialized and fossil-dependent nations

are engaging in the transition.

The case studies in India, Ethiopia, and Nigeria emphasize the centrality of electrification in reducing transport-related emissions. India demonstrates that flexible EV charging strategies can significantly mitigate grid stress while lowering emissions [17]. Ethiopia's scenarios highlight ambitious adoption of hybrid and electric vehicles (7.5–75 % by 2050), coupled with electrification of freight and rail transport [18]. Nigeria explores alternative pathways, showing that CNG-hybrids are currently the lowest-cost option, while electrification offers long-term decarbonization potential [19]. Additional evidence from local public transport (LPT) systems reinforces this trend: in Rio de Janeiro, the inclusion of e-buses and electric car-sharing combined with renewables showed significant CO₂ reductions [20], while Portugal has planned a transition to a fully zero-emission bus fleet by 2034, 83 % of which will be battery-electric [21]. Similar approaches have been studied in highly decarbonized contexts like Iceland [22], as well as in Morocco [23] and Sub-Saharan Africa, where an integrated e-bus, e-taxi, and minibus system powered by PV microgrids was shown to deliver both mobility and local energy co-benefits [24]. These examples underline that electrification of mobility is indispensable for climate targets, but its feasibility is conditioned by infrastructure readiness, affordability, and supporting policy frameworks.

Several national strategies underscore the importance of renewable energy integration to enable transport decarbonization. Ireland demonstrates a pathway to 100 % renewable electricity by 2050, reshaping system configurations [25]. Costa Rica similarly requires an additional 4.4 GW of renewable capacity to achieve an 87 % reduction in emissions compared to BAU by 2050 [11]. Germany highlights the role of sector coupling and hydrogen, noting that full electrification scenarios raise power system costs by 9–19 % but deliver substantial transport self-sufficiency [26]. European case studies further show how renewable integration supports bus electrification, such as Milan's technical-economic sensitivity analysis of e-bus deployment [27], Greece's assessment of environmental and social effects [28], and Malaga's transition strategy from ICE buses to e-buses [29]. More broadly, renewables planning and optimization tools play a pivotal role in this transition: reviews of modeling approaches [30], simulations of diesel-PV hybrid systems [31,32], and distributed generation optimization [33] all highlight that renewable-based mobility depends on cost-effective system design. Recent work on renewable-powered charging infrastructure also demonstrates the feasibility of solar-based EV charging stations [34,35], pointing to a close link between transport decarbonization and electricity system planning. Together, these studies confirm that transport electrification must proceed in tandem with renewable energy expansion and cross-sector coordination to ensure system reliability and cost-effectiveness.

Table 1 summarizes national decarbonization studies across multiple countries, detailing methodologies, scenario frameworks, and projected outcomes. Reported results include CO₂ emissions avoided, renewable energy shares, and key system impacts, highlighting diverse pathways such as electrification, EV integration, and sector coupling. Together, the cases illustrate region-specific strategies toward achieving mid-to long-term climate targets.

HOMER software is widely used for optimizing renewable energy systems in transport electrification, particularly for electric vehicle (EV) charging stations and depots. For instance Ref. [40], applied HOMER Pro to design a hybrid renewable system (solar, wind, biomass) for e-bus charging in Fez, Morocco, achieving a cost of energy (COE) of 0.08 \$/kWh but relying on grid connectivity, limiting well-to-wheel zero-emission potential. Similarly [41], used HOMER Grid to optimize an 850 kWp PV system for a Malaysian bus depot, reducing CO₂ emissions by 565 tons/year through charging routine shifts, though without operational fleet adjustments. These studies excel in cost-emission trade-offs but rarely incorporate dynamic fleet scheduling or user-centered service planning, critical for urban mobility efficiency.

LEAP (Long-range Energy Alternatives Planning) models are

Table 1
Comparative case studies on public transport decarbonization.

Country	Methodologies Used	Scenario Name	Year	CO ₂ Emissions Avoided (MtCO ₂ e)	Renewable Energy Share (%)	Key Outputs
Mexico [36]	Whole-economy DDP modeling	DDP	2050	244 (gross), 51 (net)	>55 %	Major emissions reduction across sectors; policy recommendations.
Costa Rica [11]	Integrated assessment modeling with uncertainty	Decarbonization Pathway	2050	~87 % reduction from BAU	Requires 4.4 GW additional capacity	87 % BAU emission reduction by 2050; renewable capacity increase.
Norway [37]	LEAP energy system modeling	Mitigation Scenario	2050	7.30 (mitigation), 56.40 (baseline)	Not specified	Energy demand and emission reduction; sectoral insights.
Ireland [25]	Whole-system modeling	100 % RE Pathway	2050	Not specified	100 %	Pathways to 100 % renewable energy; system configuration insights.
India [17]	Scenario-based EV charging modeling	EV Charging Flexibility	2037	Scenario-dependent	Not specified	Flexible EV charging reduces emissions; grid impact analysis.
Germany [26]	Sector coupling energy system model (GRIMSEL, open-source)	Full Electrification	2050	~132 (80 %+ reduction vs. implied baseline; average 1.3 MtCO ₂ /year avoided)	>75 (PV for hydrogen production)	Transport self-sufficiency up to 50 %; power system costs +9–19 %; battery EVs dominant
United States (New York) [38]	Machine learning (ELM, SVM-ELM, Grey Prediction Model)	Pathway 1–3 (High Tech, Integrated, Conservative)	2030–2050	18.8–57.1 (28–85 % statewide reduction; transport ~27.9–30.8 %, 2021 baseline 67.3 MtCO ₂ e)	70–100 (power sector by 2030–2040)	40–100 % statewide reduction; full light-duty EV shift by 2030; 0–10 % zero-emission vehicles by 2050
Ethiopia [18]	LEAP bottom-up, IPCC Tier 1	Maximum Ambition, NDC-Aligned, Late Action	2050	19.63 per intervention (9.68 road freight electrification, 9.95 rail expansion; total not reported)	Not reported (wind/solar/hydro for EVs)	Net-zero by 2032–2050; 7.5–75 % hybrid/EV share by 2030–2050; lowest to highest energy use
Ghana [39]	Mixed-integer linear programming (MILP) and Generation expansion planning (GEP) with unit commitment (UC)	Decarbonization Pathways	2020–2040	99–189 MtCO ₂ e	10.9–47.8 % across scenarios	Total discounted cost to meet demand is 13–17 billion US\$; cost to meet emission targets is 11–39 US\$/tonne. The study also notes that lower discount rates can lead to more long-term investment in renewable energy
Nigeria [19]	Analysis of alternative fuels/powertrains (CNG, electric)	Alternative Transition Fuels	N/A	Up to 98 % reduction vs. diesel baseline (~52 MtCO ₂ e total road transport 2020, avoided ~51 Mt implied)	Not reported	CNG hybrid lowest cost (US\$0.25/km); natural gas 52 % reduction, 33 % resource conservation

common for forecasting energy and emission scenarios. For example [42], used LEAP to project Chile's 2030 energy demand, emphasizing renewable integration but not detailing transport-specific operational impacts. Multi-criteria approaches, such as ANP, are employed to assess barriers, as seen in Ref. [43], which identified high e-bus costs and infrastructure deficits in Valencia, Spain, but did not propose integrated renewable solutions. Life Cycle Assessment (LCA) studies, like those by Ref. [44], highlight the environmental benefits of EVs but often overlook depot-level renewable integration, focusing instead on vehicle production emissions.

Reviewed studies highlight electrification, renewable energy expansion, and EV adoption as key decarbonization enablers, yet significant gaps persist. In developing countries like Ethiopia, Ghana, and Nigeria, analyses often lack detailed renewable integration pathways for transport, with methodologies varying widely, hindering cross-country comparisons [18,23,45]. Few studies address socio-economic dimensions such as equity, affordability, or co-benefits like air quality, despite evidence of public health gains from urban decarbonization [9, 24]. Mediterranean city research, such as Valencia's focus on electrification barriers, rarely integrates operational fleet rescheduling with depot-specific renewable systems [43]. While HOMER-based studies like Fez, Morocco, optimize PV-based charging (e.g., 280,610 tons CO₂/year avoided), they prioritize economic-emission trade-offs over user-centered service planning or depot siting [23]. This study addresses these gaps with a scalable methodology integrating multi-scenario fleet rescheduling (e.g., Bus Rapid Transit and Light Rail Vehicles with 5–20 min frequencies, 15–25 km/h speeds) and HOMER-optimized PV microgrids (12.143 MW and 8.873 MW) with Li-ion storage. Applied in Alexandria, Egypt, it achieves 1.617 GWh/month supply, 17,544 tons CO₂/year savings, and \$1.4M in avoided emission costs, offering a

holistic framework for net-zero urban transport transitions.

1.2. Contributions

This study addresses critical gaps in public transport decarbonization by proposing a scalable methodology that integrates user-centered fleet rescheduling with renewable-powered depots, achieving well-to-wheel zero emissions. Unlike prior studies focusing on isolated charging optimization (e.g., Morocco's e-bus depots [23]) or broad energy scenarios (e.g., Ethiopia [18]), our approach holistically combines operational and energy system design [9,23,24,43,45]. The methodology includes.

- 1. Fleet Rescheduling:** Analyzes local public transport (LPT) routes to design electrified systems using e-buses and trams (Bus Rapid Transit and Light Rail Vehicles). Simulations vary commercial speeds (15–25 km/h) and frequencies (5–20 min) to optimize stops, fleet size, and monthly kilometers, ensuring service continuity and user satisfaction.
- 2. Energy Demand Modeling:** Assigns consumption coefficients (kWh/km) to vehicle types, estimating total electric demand across scenarios based on scheduling and speeds.
- 3. PV Microgrid Optimization:** Uses HOMER to size PV-powered microgrids and Li-ion storage, tailored to depot-specific load profiles, ensuring uninterrupted power supply [23].
- 4. Depot Siting:** Selects optimal depot locations based on proximity to LPT lines and available rooftop area, enhancing technical feasibility.
- 5. Scalability and Replicability:** Designs a modular framework adaptable to diverse urban contexts, particularly Mediterranean and developing cities.

6. Sensitivity Analysis: A multi-scenario sensitivity analysis links operational demand growth (+1 %, +3 %, +5 %) to techno-economic, environmental, and social outcomes, showing how PV capacity, storage, costs, and CO₂ savings adapt across trajectories, thereby confirming the methodology's resilience and scalability.

Applied in Alexandria, Egypt, the methodology electrifies LPT with two PV-integrated depots (12.143 MW and 8.873 MW, covering 1.09 km² and 0.126 km²), producing 1.617 GWh/month at 0.418 \$/kWh. This achieves 17,544 tons CO₂/year savings and \$1.4M in avoided emission costs, with excess energy for resale or other utilities. Unlike fragmented approaches [18,23], this study ensures user-centric service (10-min frequency), reduces congestion, and supports net-zero transitions, offering a replicable model for sustainable urban mobility.

1.3. Paper organization

This paper is structured into six sections to systematically address public transport decarbonization. Section 1 introduces the decarbonization challenge and reviews global strategies and tools for renewable integration in transport, identifying gaps in holistic fleet-renewable-depot models. Section 2 outlines the proposed methodology, which encompasses user-centered fleet rescheduling and PV-powered depot design with energy storage. Section 3 presents the Alexandria, Egypt case study, applying the proposed method to electrified transport and depot integration. Section 4 evaluates results and discusses operational outcomes and PV microgrid performance, including feasibility and excess energy potential. Section 5 discusses the results by providing sensitivity analysis. Section 6 concludes by emphasizing the methodology's scalability, its contributions to net-zero transitions, and its advantages alongside its limitations.

2. Proposed methodology

In this section of the study, the methodology employed to transition from the current mobility service to an integrated and innovative mobility service, centered on fully electric vehicles, which can encourage citizens to utilize the LPT, will be outlined. Additionally, the

objective is to integrate a transportation system that minimizes GHG emissions and ensures sustainability through the supply of green energy. The first step of the methodology involves measuring the length of the identified route by aggregating the kilometres travelled daily during service and off-service periods. Subsequently, an energy consumption factor will be determined, evaluated in kWh/km, and combined with the kilometres travelled. This process will enable the determination of the daily energy required to power the entire fleet. In the case study, inherent limitations of electric vehicles, such as range, also need to be identified. Based on the collected energy consumption data, the results will be extrapolated to assess the monthly energy consumption, considering energy storage production. In this study, PV panels will be utilized to supply green energy for powering the fleet whenever possible. Conversely, if utilizing PV energy directly to power the fleet is unfeasible, it will be stored in stationary electrochemical storage systems. Fig. 1 provides a compact, high-level view of the study context, linking the redesign of the public transport service to the energy demand that derives depot electrification. The diagram synthesises route configuration, headways and commercial speeds, and depot siting near line terminals, from which fleet sizing and daily distance run are derived. These operational choices determine the daily energy demand and thus set the boundary conditions for the subsequent microgrid design and evaluation.

Fig. 2 details the schedule-aware methodological pipeline from transport service design to techno-economic and environmental evaluation. The process begins with route geometry and timing parameters, which feed into the renewed mobility service module. This stage establishes the commercial schedule, quantifies the number of vehicles required, computes the daily distance travelled, and ultimately determines the daily energy demand. The resulting demand profile, combined with the selected load characteristics, drives the optimal microgrid sizing stage. Here, photovoltaic generation and energy storage capacities are independently optimized to support the transport service. The optimized configuration then flows into two key evaluation branches. The economics module estimates net present cost and cost of energy, while the social impact module quantifies avoided CO₂ emissions and their associated monetary benefits. A sensitivity analysis enhances the assessment through variations in daily load and long-term

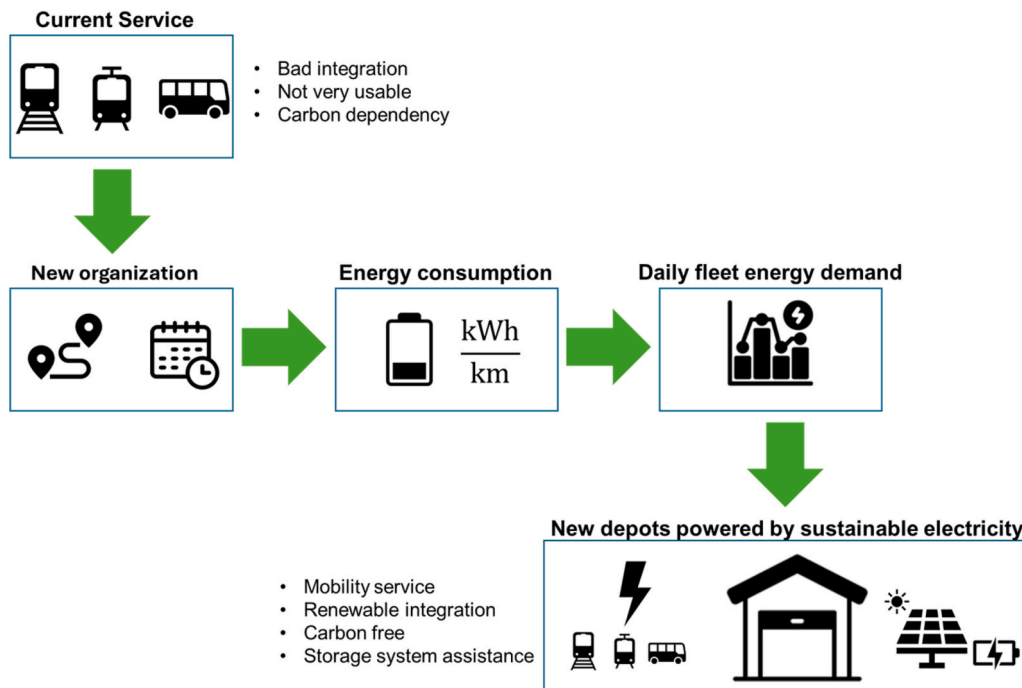


Fig. 1. Sustainable mobility service reshaping, high-level process from transport service design to electric depot based on microgrid implementation.

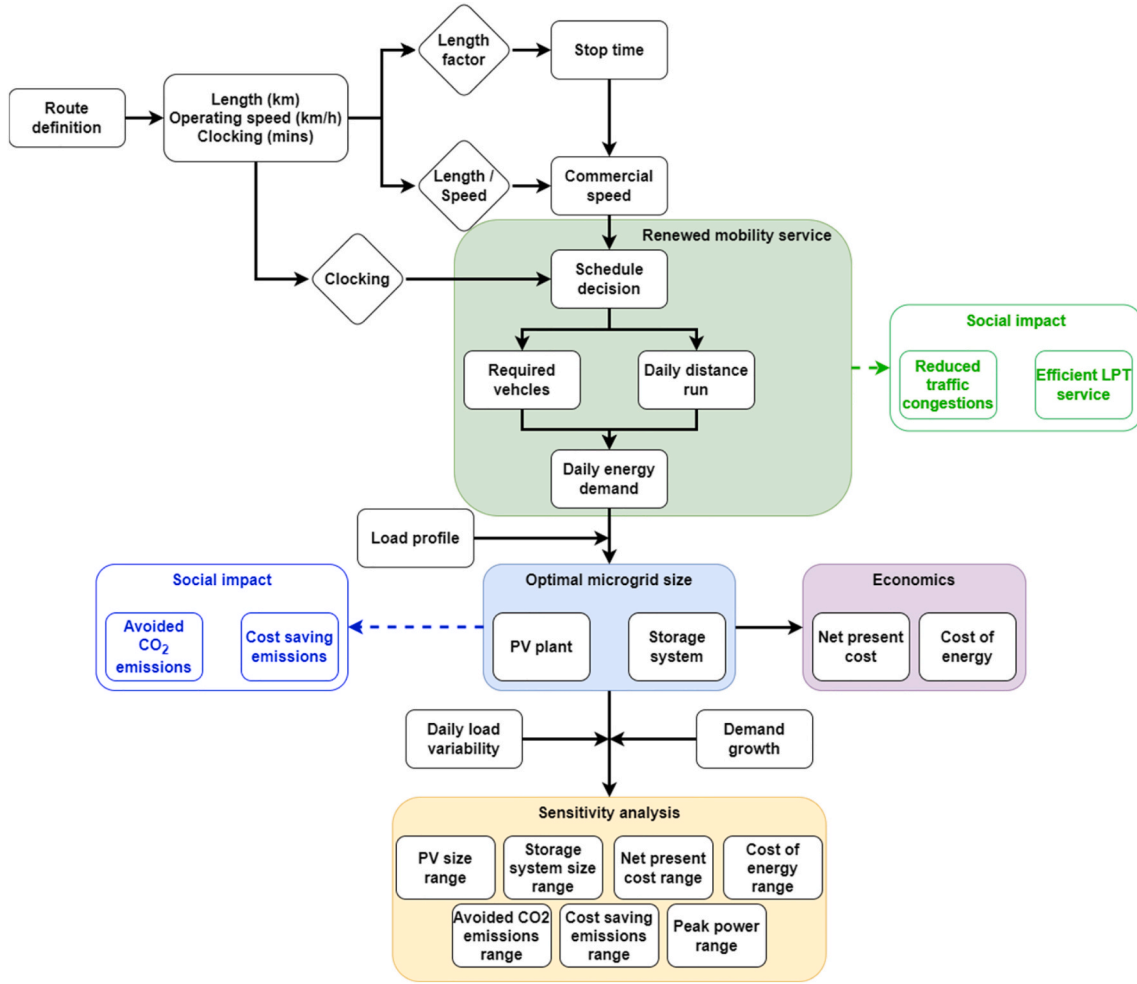


Fig. 2. Sustainable mobility service reshaping, methodological flowchart from transport service design to techno-economic and environmental evaluation.

demand growth, yielding performance ranges that reveal cost-emission trade-offs and demonstrate the robustness of the proposed design.

This work contributes to the field by advancing prior Homer-based analyses in three ways. Initially, the electric depot load is derived from the actual service schedule. This process involves the conversion of headways, layover times, and depot availability into an hourly charging demand and the associated peak constraints. This schedule-aware pipeline preserves the operational logic of public transportation, such as simultaneous evening returns and morning pull-outs, which is typically lost when using generic load shapes. Secondly, the schedule-based demand is coupled with a multi scenario analysis, with different demand growth, reporting not only the PV and BESS sizing but also NPC, CoE, peak load and load factor, thus linking operational drivers to techno-economics. Finally, transparency and reproducibility is sustained by the equations, inputs and descriptions presented in this section, keeping clear the presence of limitations in the approach.

2.1. Mobility service renewal

The definition of routes is contingent upon the placement and interconnection of two interchange points along a single path. In the context of Mediterranean cities, parallel routes are predominantly utilized in local public transportation due to the movement of residents along the coastlines. Consequently, this methodology will be applied to the primary routes that follow this topography for various modes of land transport. For BRT, it is essential to delineate the main routes. Subsequently, the length of each route (L) must be measured, identifying the

maximum number of stations N_{maxst} to consider for each path, calculated using equation (1):

$$N_{maxst} = L \times f_l \times f_p \quad (1)$$

Service planning involves two phases: speed control and scheduling. Speed control sets trip duration using running or commercial speed, with 30-s stops for passenger boarding/alighting. Scheduling sets BRT intervals at 5, 10, or 15 min and LRV intervals at 10 or 20 min, aligning trip starts with divisible intervals (e.g., 5-min intervals start on 5-min marks). Out-of-service depot travel distance is calculated, with a 200 km/charge threshold; exceeding this splits trips across multiple buses to meet service demands. This process enables the determination of the minimum number of buses required to conduct daily BRT service, as outlined in equation (2). Conversely, LRT service employs vehicles continually powered via a pantograph, with the number of vehicles dictated solely by the service schedule.

$$Max_{BRT} = \min \left\{ \frac{L}{N} \leq 200 \right\}; N \in \mathbb{Z} \quad (2)$$

As mentioned, the distance travelled by buses will be multiplied by a coefficient that identifies the energy consumption to determine the daily electrical demand of each bus. Consequently, the energy required by the buses will be aggregated to identify the total energy demand of the fleet. The coefficient identified through experimental trials for 18 m buses is 1.75 kWh/km [46]. The vehicles for the LRV will use an experimental coefficient of 1.55 kWh/km [47,48].

In order to identify an efficient LPT system, it is necessary to identify

the points of greatest turnout and interchange of passengers. These points represent nodes in the LPT network, and correspond to the positions of mobility hubs. In addition to these points, it is important to identify the optimal location of vehicle storage and maintenance facilities. If the current depot used for the tram is an adequate structure, the bus depot will not be adequate, therefore the implementation of a new depot will be necessary.

2.2. Depot electrification methodology

Electric mobility offers efficient, emission-free transport but requires sustainable energy sources to ensure environmental benefits. This citizen-centered study enhances Alexandria's public transport by integrating renewable-powered depots. Two depots, treated as one for aggregate power demand sizing, support high-power on-route charging (upside-down pantograph) or overnight plug-in EVSE charging. A PV microgrid, sized using HOMER software, meets fleet demand, with production scaled by depot size for PV panel distribution. This modular, off-grid-capable solution drives decarbonization and supports sustainable urban mobility. Moreover, this technology can be seamlessly integrated with RES [49]. Finally, through simulation, it will be possible to determine the optimal configuration of the system. Verification will be conducted by comparing the utilization rates of the components. Specifically, it will be necessary to ensure that the total power generated by the PV plant throughout the year is sufficient to meet the demand, taking into account the usage of the (Battery Storage System) BSS. In general, it must be ensured that the combined power supplied by the generating elements (P_{grid} , for the power injected from the grid and P_{PV} , for power supplied by PV panels) and the storage elements ($\sum_{i=1}^n P_{storage_i}$) always exceeds the power required by the load (P_{load}) combined with losses (P_{loss}), as verified by equation (3).

$$P_{grid} + P_{PV} \pm \sum_{i=1}^n P_{storage_i} - P_{loss} - P_{load} = 0 \quad (3)$$

This study presents several limitations that should be acknowledged. First, the load modelling approach does not fully capture the temporal variability of passenger demand and service frequency in public

transport. However, the model followed the existing vehicular schedule, making it reliable in ordinary service situations. Second, battery degradation is considered through a simplified model only for maintenance and replacement, without considering the performances degradation. Then, stochastic variability of the solar resource is not included, which could influence reliability and COE. However, weather data is based on the history of the database used (US Renewable Resource Data Center and NASA), increasing performance reliability [50]. Finally, the analysis is performed in HOMER, a widely adopted optimization tool, since the reliability of the internal algorithms. To improve the transparency and reproducibility of the case study, all key input parameters, equations for the economic and environmental indicators, and explicit assumptions are reported. Moreover, the adoption of HOMER ensures comparability with numerous prior studies that use the same tool [51–53].

2.2.1. Load sizing methodology

The first step in electrifying the depot involves sizing the load model. The electrical load is calculated based on the transportation service offered by the PTO. Designing the off-grid support system requires careful selection of components to ensure reliability and efficiency. The daily load profile model used for simulation can be chosen from four different models, each varying based on characteristics, as depicted in Fig. 3.

Commercial load (Fig. 3a) exhibits significant variability depending on the evaluated day (weekday or weekend). Consumption is minimal during the night hours (9 p.m.-6 a.m.) and rises during the day (7 a.m.-8 p.m.), reflecting the consumption pattern of commercial activities. Community load (Fig. 3b) demonstrates reduced variability depending on the day (weekday or weekend). Demand is lowest in the early hours of the day, gradually increasing until the evening (9 p.m.), before declining again. Considering the operational behaviour of a depot, which needs to recharge a limited number of vehicles during the day while meeting office requirements, and faces higher load at night due to more buses recharging, the Community model appears most suitable for determining the electrical load. Industrial load, as depicted in Fig. 3c, shows minimal variability between weekdays and weekends. This load model characterizes large industries that cannot suspend operations.

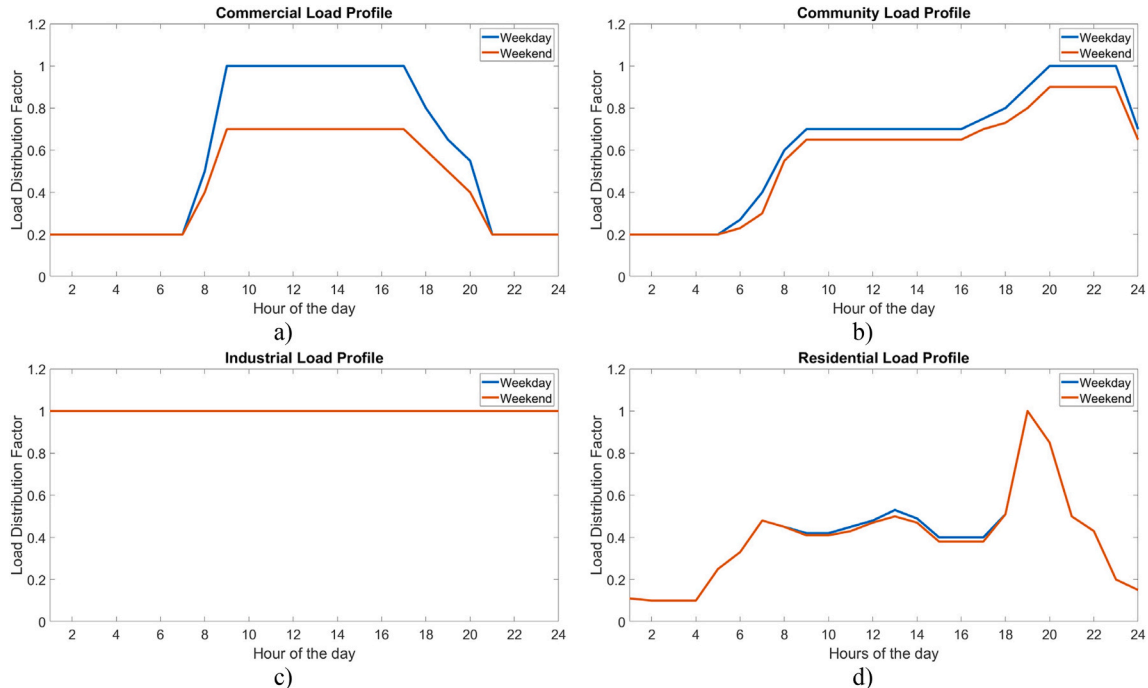


Fig. 3. Daily load profile models: a) Commercial load profile, b) Community load profile, c) Industrial load profile, d) Residential load profile.

Residential load, illustrated in Fig. 3d, exhibits limited variability between weekdays and weekends. This model, as implied by its name, is well-suited for planning residential loads, particularly given the load variability throughout the day. The input variable of the system will thus be the electrical load demand, identified through the methodology outlined in Section 3.1.

2.2.2. PV sizing methodology

To comply with (3), an adequate power source must be implemented. In this work, a PV system has been selected to power the microgrid, enabling the depots to be sustainably powered. The selected plates are high efficiency PV panels, mounted on the rooftop of the building. The PV plates considered have a rated power of 1 kW and the size is 6 m² [54]. To evaluate the expected photovoltaic production is used (4) [55]:

$$P_{PV}(t) = \frac{I_g(t) \cdot P_{PV}^n \cdot \eta_{pan}(t) \cdot \eta_{inv} \omega_{deg}(t)}{I_{STD}} \quad (4)$$

The plates ensure sustainable electricity production, which will be generated during the peak solar radiation moments of the day. For the sizing of the system, knowledge of the daily GHI is required to define productivity in kWh/m²/day. Once these data are collected and combined with the characteristics of the PV panel, equation (4) allows for the definition of the optimal size of the plant in terms of rated power in kW. This information enables the identification of a portion of the electric output of the system. At the same time, knowing the daylight hours makes it possible to map the annual production of the PV system's panels.

A charge controller (CC) with MPPT regulates battery energy flow, preventing overcharging or deep discharges to ensure longevity. Inverters convert DC to AC, with size and type based on energy demands and maximum DC input voltage. A combined solar-battery inverter, with 95 % efficiency and a 15-year lifespan, is selected to meet system requirements, supporting sustainable power for Alexandria's electrified public transport depots.

Finally, when multiple PV plant is considered, the total installed PV capacity (P_{tot}) is allocated proportionally to the available surface area of each PV allocation depot. Denoting with $A_{depot-i}$, the area of the i^{th} depot and with n the number of depots, the installed PV capacity assigned to each depot is computed through (5).

$$P_{depot-i} = \frac{A_{depot-i}}{\sum_{i=1}^n A_{depot-i}} \cdot P_{tot} \quad (5)$$

This formulation ensures that the PV sizing remains consistent with the real spatial constraints of the depots, while keeping the overall installed capacity equal to the system-level optimum derived from the simulation.

2.2.3. Storage system sizing methodology

A BSS is implemented to store surplus energy generated by the PV plant for later use. The selection of batteries is critical to meet energy demand and ensure network stability. The chosen storage system consists of 1 MWh Li-ion batteries. These batteries are designed with a Depth of Discharge (DoD) of 80 % to prolong their useful life. The nominal voltage is 600 V, and the nominal capacity is 1666.7 Ah, with a roundtrip efficiency of 90 %. In the model utilized, the system control variable is the State of Charge (SoC) of the BSS. The SoC must never fall below a minimum threshold value, which is determined by equation (6) [56]:

$$SoC_{t_1} = SoC_{t_0} - \int_{t_0}^{t_1} \frac{i_{BSS}(t)}{C} dt \quad (6)$$

The storage system is combined with the PV panels and the CC system, which regulates the charging and discharging cycle of the batteries. Knowing the battery inputs, the load demand, the characteristics of the

PV system, and the CC, equations (3) and (6) are used to size the storage system. The incorporation of a BSS entails adjusting the generated electrical output. Moreover, through equation (6), it is possible to monitor the utilization cycle of the BSS throughout the year, highlighting the periods when they will be under the most stress. Finally, (6) is implemented in a discrete time domain with an hourly time step, consistent with the temporal resolution of the software simulation. The storage charging-discharging efficiency is assumed constant at 90 % in line with the case base assumptions. Therefore, battery degradation effects, such as cycle aging, are not modeled explicitly; instead, replacements are accounted for the cost analysis. Therefore, it is considered that degradation is not affecting performances during the 15 years of operation before replacement, this is consistent with existing theoretical studies [57,58]. This simplifying assumption is consistent with existing techno-economic studies on PV-battery systems, where long-term performance degradation is either neglected or addressed only through replacement schedules [59,60]. Similarly, the stochastic variability of the system parameters is not embedded in (6), but handled through deterministic average values, which is common practice in renewable microgrid modelling [61].

2.2.4. Cost analysis methodology

Finally, it is possible to calculate the Net Present Cost (NPC) of the elements considered to define a cost analysis of the micro-grid. Equation (7) evaluates C_{NPC} by considering the sum of costs arising from installation (c_{in}), operational (c_{op}) and fuel (c_{fuel}), reduced by the present values of the components (V_{pres}) and the value of revenues (V_{Rev}).

$$NPC = (c_{in} + c_{op} + c_{fuel}) - (V_{pres} + V_{Rev}) \quad (7)$$

Furthermore, the methodology allows to calculate the produced energy cost from the plant. Through the NPC it is possible also to evaluate the annualized costs, that is the cost to sustain to have the same NPC, but divided year by year. This value is relevant since it impacts on the leveled Cost of Energy (CoE) produced by the system as reported in equation (8):

$$CoE = \frac{C_{ann_{tot}}}{E_{served}} \quad (8)$$

The economic evaluation proposed by (7) and (8) follows the framework of the simulation software. All values are expressed in real terms, assuming a constant-currency basis for the year 2024. A real discount rate of 6 % is adopted in the study, consistent with recent studies. The project lifetime is set at 25 years, in compliance with PV modules lifetime [62]. The investment cost for the PV system was fixed at 3000 \$/kW. Over project lifetime, replacements of components with shorter life expectancy were scheduled, and the salvage value of assets with residual lifetime was credited at the end of year 25.

It is useful to quantitatively assess the social and economic impact that follows from the methodology adopted. Specifically, one wants to economically assess the value of CO₂ avoided per year. The previous stages of the methodology have already helped to establish the annual energy produced by the microgrid. In this step, using equation (9), the energy used was combined with the average grid emission factor (f_{CO_2}) to compute the avoided CO₂ emissions per year [63].

$$ay_{CO_2} = E_{served} \cdot f_{CO_2} \quad (9)$$

Once the avoided CO₂ emissions per year have been identified, it is possible to identify a factor that determines the cost per ton of CO₂ emitted (P_{CO_2}). Following [64], 80 \$/CO₂tons is considered. Through equation (10), it is possible to identify the CO₂ Saving Value (CSV), which economically quantifies the impact of the CO₂ saved due to the microgrid.

$$CSV = ay_{CO_2} \cdot P_{CO_2} \quad (10)$$

3. Case study: Alexandria

Alexandria, Egypt's second-largest city (5.5M inhabitants), faces mobility challenges due to heavy reliance on private cars, causing traffic congestion and poor air quality in its sub-tropical climate. The current public transport system includes 738 buses (some e-buses since 2020), a 100 km tram network with 141 vehicles, and a 22 km urban railway with 6 vehicles. This study proposes replacing buses with electric BRT and trams with modern LRVs. Following Section 2's methodology, we reorganize local public transport by defining new stops and simulating scenarios with varying commercial speeds (15–25 km/h) and frequencies (5–20 min) to calculate fleet size and energy demand. A PV-powered microgrid with storage, sized via HOMER, will supply depots, ensuring sustainable charging and supporting net-zero transitions. In Fig. 4 is reported the Alexandria Governorate.

3.1. Alexandrian LPT organization

Considering the coastal morphology of the urban area of Alexandria, which extends along the coast, three parallel lanes will be designated for the BRT service to represent the three main routes that enhance accessibility to Alexandria for passengers. Figure A1a depicts the new bus lines that will traverse the city, while Figure A1b shows the existing tramlines in blue and red, with the anticipated additional tramlines indicated in yellow and green. Table 2 highlights the terminals of the lines, their respective lengths, and the number of stations derived from the applied methodology.

New BRT lines, integrated with existing and upcoming LRVs and urban railway, strengthen Alexandria's transport network using the proposed methodology. A multi-scenario analysis evaluates commercial speeds (10–25 km/h), frequencies (5–20 min), and vehicle consumption to estimate daily and monthly distances, energy demand, and minimum fleet size. This informs PV system sizing for depots, ensuring sustainable power supply for bus lines while maintaining service efficiency. Table A1 and Table A2 display the monthly kilometres served, the energy demand of the vehicles, and the minimum number of vehicles needed to operate the lines. Different operating speeds of the vehicles and the spacing between vehicles are considered, which are crucial for determining the vehicles required to serve the lines.

Nine BRT and six LRV scenarios were analyzed, assessing kilometers served, energy demand, and minimum daily fleet size to inform PV microgrid sizing. Scenarios vary commercial speeds (10–25 km/h) and frequencies (5–20 min). The PTO favors higher speeds (e.g., 20 km/h) to minimize vehicles and costs, while users prefer frequent service (e.g., 10-min intervals). Optimal scenarios (20 km/h, 10-min frequency for BRT and LRV, per Table A1 and Table A2) balance efficiency and user satisfaction, avoiding delays and overcrowding. These inform energy demand and fleet requirements for sustainable depot powering. Considering the uniform consumption of the vehicles on a monthly



Fig. 4. Alexandria Governorate.

Table 2

Routes description.

Route	Terminals	Length	Stations number
BRT 1	Abu Qir – Ras Tin	25.7 km	76
BRT 2	Al Mandara - Misr	15.2 km	31
BRT 3	Al Ma'mora – Al Maks	25.4 km	51
LRV 1*	Victoria – Al Raml	10.5 km	31
LRV 2*	Victoria – Al Raml	10.2 km	29
LRV 3	Al Raml – Karmouz	10 km	29
LRV 4	Moharam Bek – Al Maks	11.2 km	33

basis, it is possible to determine that the annual consumption of the two services will be those reported in Table 3.

Finally, to complete the selected mobility system, new depots will be implemented capable of accommodating the vehicles that will be used in the service. The choice of the depot location was based on its proximity to the three main routes of the BRT service, as having a separate depot for each line would be inefficient and costly. Considering accessibility, available space, safety risk factors, the optimal depot location can be identified by evaluating potential expenses. The location must be accessible and minimize the out of service distances covered to maximize actual service. The size must be sufficient to accommodate all vehicles from the routes and depot facilities necessary for vehicle maintenance and charging, as well as parking, offices, training facilities, and security structures. Finally, it should be noted that the implementation of a depot could be a significant investment, so it is advisable to consider budget-compliant locations. In this work, the selection will primarily be based on the depot's surface area and proximity to the line terminals. Therefore, two areas in the city of Alexandria have been identified that may be suitable for the construction of the new depots and are shown in Fig. 5

The identified depots are in the two opposite poles of the city, and near the line terminals, so as to minimize energy consumption to return to the depots. The areas identified are those of Al Ma'mura and Al Qabari. The depot in the Al Ma'mura area has a surface area of approximately 1.09 km², while the depot located in the Al Qabari area has a surface area of approximately 0.8 km². The two depots will be electrified to accommodate the vehicles and a PV plant will be implemented to meet energy demand.

3.2. Depot electrification simulation

Taking into account the load model in Fig. 3, each of the load profiles is made up of a series of load distribution factors, which are used to identify the quantity of electrical load for each specific hour of the day. Considering the behaviour of a depot, which during the day would have to recharge a reduced quantity of vehicles and satisfy the needs of the offices, while at night it would have a higher load since more buses are recharging, the Community model seems to be the most suitable for building the electrical load. Following the characteristics of the mobility system and the charging requirements which are larger during the evening, when the mobility service is reduced, the charging model that will be used in this case study for depots is the community one. Considering the monthly LRT and BRT combined load of 1.617 GWh, the average daily load is 53.808 MWh/day. This value is spread during the day following the community load profile. Moreover, it is considered an hourly load variability of 20 % and a daily load variability of 10 %. By

Table 3

Energy consumption [GWh] of the BRT and LRV services implemented.

Service	Monthly energy required [GWh]
BRT	1.053
LRV	0.564
Total	1.617



Fig. 5. Suitable area selected by lines proximity and space availability for new RES integrated depots.

adjusting the load variability, it's possible to achieve various configurations of the microgrid. Specifically, the number of batteries utilized, and the quantity of PV plates will vary, as the microgrid is sized based on the electrical demand of the depots. The peak load for the sized model is 6.671 MW and this brings a load of 34 %, that is evaluated as the average load divided by the peak load. Once the load to be powered has been identified, the model requires the inclusion of the systems to be used in the microgrid. Fig. 6 represents the microgrid model used for the implementation of the aggregate electrified depot as a load.

In the microgrid, PV panels with an area of $6 \text{ m}^2/\text{kW}$ are used. Each of the simulated PV panels will have a rated power of 1 kW. The software calculates the average annual production in $\text{kWh}/\text{m}^2/\text{day}$ based on the location and assumes a cost of 3000 \$/kW for the installation of each PV panel. This will allow for the calculation of the microgrid cost and the CoE produced with these panels based on the investment. The PV panels have a considered lifespan of 25 years. Finally, the derating factor is also evaluated. The derating factor adjusts the power output of the PV array to reflect reduced performance in real-world conditions compared to its rated conditions. It considers factors such as panel soiling, wiring losses, shading, snow cover and aging. In this case study a derating factor of 80

% is used. Photovoltaic panels are supplemented with a BSS composed of Li-ion batteries to ensure system stability and the ability to serve the load even during nighttime hours. The system utilizes batteries with the same nominal voltage as the DC Bus, thus they are connected in parallel. These batteries have a nominal capacity of 1 MWh. However, the effective capacity that the batteries can deliver is determined by setting a maximum DoD to avoid excessive degradation. Considering an acceptable DoD of 80 %, this translates to 800 kWh of availability for each battery unit. The round-trip efficiency of the battery represents the energy efficiency of the complete charging and discharging cycle, indicating the fraction of energy input into storage that can be subsequently retrieved. It is assumed that the charging and discharging efficiency of the battery are both equal to the square root of the round-trip efficiency. Finally, a high-efficiency bi-directional electric converter is incorporated to facilitate communication between the DC bus, where the BSS is located, and the AC bus, where the load and PV system are situated. Following all these characteristics, it is possible to size the depot PV plant to size in the new depot.

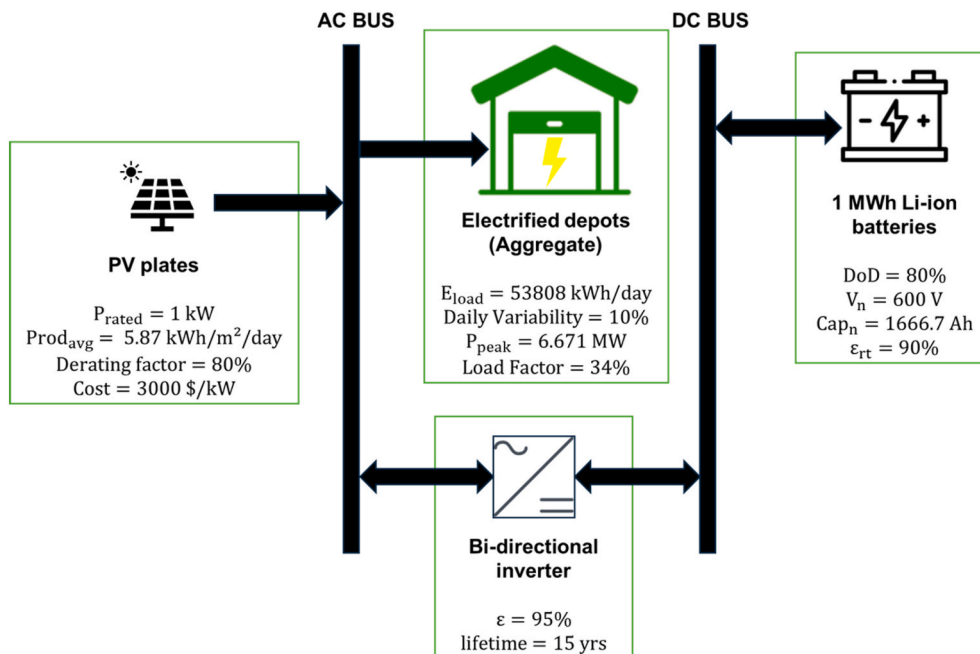


Fig. 6. Micro-grid structure of a PV integrated depot sided by a storage system.

4. Baseline simulation results

Once the service has been implemented, both the service demand and energy demand are identified. The identified monthly consumptions are then distributed on a daily and hourly basis, utilizing the load models identified in Fig. 3. The load model employed in the simulation is the community load profile. The selection of this load profile reflects the usage pattern of the vehicles, which primarily charge during nighttime hours. Moreover, the model exhibits higher loads on weekdays compared to weekends, aligning with the vehicle management practices of a PTO. Subsequently, the simulation is initiated, enabling the sizing of a PV plant adequate to meet the energy demand using dedicated software, which relies on the combination of (4) and (6) to generate the appropriate output. The monthly production of the PV system is then determined, as depicted in Fig. 7, which illustrates the monthly power produced by the PV system. It is evident from Fig. 7 that the monthly production significantly exceeds the monthly demand, even when considering a load variability of 10 %, as in the case study. However, scaling down the size of the PV plant compared to the simulated size may not ensure compliance with daily production constraints. This constraint becomes particularly stringent during the winter months and must adhere to equation (3). Therefore, following this initial analysis of the monthly output, more precise and in-depth assessments will be conducted on the outputs, also from a daily perspective, utilizing heatmaps.

As expected, the PV power output respects the climatic conditions of the city; during hot and torrid summers, production is at its maximum, while in humid and rainy winters, power production is lower. In addition to highlighting the seasonal variability of PV generation, Fig. 7 provide useful insight into the relationship between supply and demand. The monthly PV production curve not only exceeds the constant depot demand across all months, but also creates a seasonal surplus margin that is widest in summer, up to +3.5 MW above power demand, and narrower in winter, +2 MW. This indicates that while the system ensures adequacy throughout the year, the balance between production and consumption is not constant; specifically, during summer months, excess PV generation could reduce the need for storage or potentially be exported to the grid, while in winter months the smaller margin suggests that storage plays a more critical role in maintaining reliability. Therefore, Fig. 7 not only confirms the adequacy of the PV sizing, but also emphasizes the importance of integrating seasonal flexibility strategies into the system. It is then important to underline the electrical outputs of the simulation, which are reported in Table 4.

The simulated plant will have a rated capacity of 21.017 MW. Considering the solar panels utilized by the system, with a power output of $6 \text{ m}^2/\text{kW}$, it can be estimated that the plant's size will be approximately 0.126 km^2 , making it perfectly suitable for the new identified locations. In addition to the rated capacity, the mean power output is identified, representing the average power generated by the plant during daylight hours. This is calculated using equation (4) and is divided by the daily sunlight hours. The mean power output of the plant is found

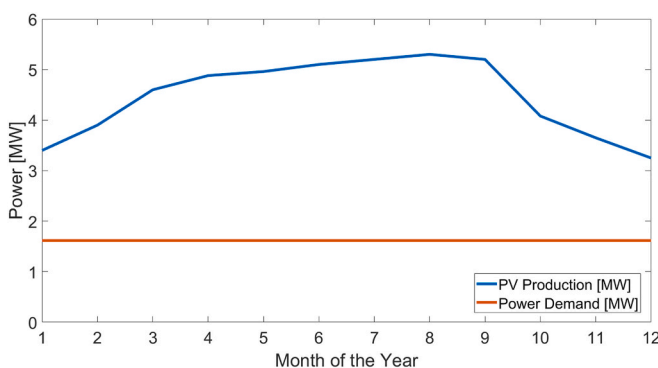


Fig. 7. Monthly PV power output produced in relation with power demand.

Table 4

Electrical output of the simulation of the overall equivalent microgrid.

Quantity	Value
Rated capacity	21.017 MW
Mean power output	2.280 MW
Mean daily energy output	107.529 MWh/day
Capacity factor	21.32 %
Yearly energy production	39.248 GWh/year
Maximum power output	21.094 MW
Yearly hours of operation	4385 h/year
Levelized cost of the production	0.102 \$/kWh

to be 2.280 MW. It is evident that the mean daily energy output represents the average energy produced by the system during daylight hours. Subsequently, the capacity factor is calculated. This is determined by dividing the total energy actually generated by the photovoltaic system during the reference period by the maximum theoretical energy that could be produced if the system operated at maximum capacity for the same period of time. This value provides an indication of the percentage of efficiency with which the photovoltaic system utilizes its maximum capacity over time. The simulation also presents the energy produced by the plant per year. This data is known thanks to the yearly hours of operation, evaluated based on the daily sunlight hours, as highlighted in Fig. 8. The maximum power output of the system is also presented. This data refers to the maximum power actually produced by the system under specific conditions. The maximum power output can vary over time and may be lower than the rated capacity of the system, which instead is the specific quantity for which the system is designed to generate. Furthermore, it is clear that annual energy production exceeds demand, despite the variability. The excess energy can however be accumulated in the BSSs in storage, to be used for overnight charging or to be reinjected into the network and sold. The potential for generating additional revenues through the sale of excess electricity is presented as theoretical scenario, with the objective of assessing the sensitivity of the model to varying boundary conditions. A thorough examination of the local electricity market prices, feed-in tariffs, and regulatory framework in the area was precluded by the lack of data available for the analysis. This investigation is therefore relegated to future research endeavours. This simplification does not impact the reliability of the study's core results, which are derived from system parameters. The use of these panels with a capacity factor of 21.32 % allows the production of useable energy at a low price, equal to 0.102 \$/kWh. Fig. 8 highlights the operation of the PV system during the year.

Fig. 8 provides a heatmap of PV productivity across the year, making it possible to identify both daily and seasonal patterns. As expected, the highest productivity is concentrated in the central months of the year, between May and August, with values steadily above the 5 MWh/day, while during winter months, in December and January the lowest outputs are shown, at 3.5 MWh/day. The heatmap also highlights a smooth intra day variation, reflecting the stability of solar irradiance in Alexandria and confirming the suitability of PV for supply. From a planning perspective, these results suggest that seasonal complementarity with storage system is essential. Summer surpluses could be exploited to reduce charging costs or export electricity, while the winter period requires careful scheduling of charging and storage management to ensure reliability. Fig. 8, combined with Fig. 7 are showing not only averages

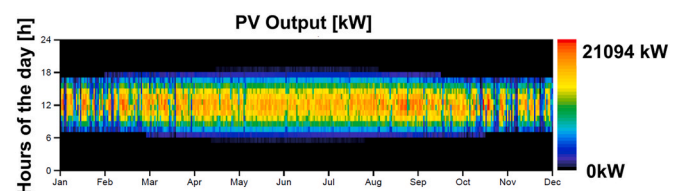


Fig. 8. PV productivity heatmap during the year.

monthly, but also the temporal distribution of PV production, which represents a key operation for depot sizing. This also provides valuable insights into the fluctuations in solar energy production based on weather conditions and the angle of the sun. These data are essential for the design and optimization of photovoltaic systems, allowing for the assessment of the effectiveness and profitability of the solar installation under various environmental conditions. The identified configuration is optimal if combined with a storage system, since in some periods of the year the PV plant's production may not be sufficient due to the weather conditions. Furthermore, the energy produced is not all used immediately as some EVs are recharged during the night. It will be worth of interest to verify the optimal size of the system with a heatmap linked to the use of the BSS. It is foreseeable that in the winter months between October and April the batteries will be discharged much more intensely, due to the lack of light during the day and the BSS will reach SoC close to the minimum threshold imposed by the system (SoC = 20 %). In addition to the PV system, a storage bank composed of 1 MWh Li-Ion batteries has been added to avoid network instability and use the micro-grid even when the PV panels are not producing. In the scenario considered, the bus voltage of the DC bus is the same at which the BSS works: 600V. 80 strings are connected in parallel to the DC bus, each containing a single storage system. The characteristics of the BSS are shown in Table 5.

Table 5 displays the nominal capacity of the batteries that is useable, excluding the 20 % of charge state deemed unusable to prevent excessive battery degradation resulting from suboptimal power throughput management. Additionally, it's noteworthy that the batteries can provide an extra day of charging autonomy. The annual throughput is also determined, reflecting the energy flow through the batteries, crucial for SoC and State of Health (SoH) management of the Battery Storage System BSS. Furthermore, the analysis considers the energy-dependent cost associated with battery wear and the annual energy throughput. This cost factor accommodates potential battery capacity loss over time due to wear and the possibility of premature replacement before the expected end of the battery's useful life. The importance of precise storage system management is underscored by the heatmap depicted in Fig. 9, where occurrences of critical SoCs approaching the lower threshold are infrequent during winter months, albeit without reaching critical values.

It is worth to notice, that the BSS SoC is very high throughout the year, with only a few per year where it approaches 20 %. This has a significant impact on the size of the storage bank and the cost of the system. This result, however, is linked to the fact that the depot, having space for PV plates and batteries, is designed as self-sufficient and powered only by the PV plant. Therefore, to meet demand even on days when PV production is minimal, the BSS will work and need sufficient stored energy. This, combined with the charge controller, which prevents the BSS from having an SoC lower than 20 %, impacts as shown in Fig. 9. The management of the SoC and PV production allows compliance with (7), validating the optimal sizing of the system. Moreover, the seasonal effects are clearly represented in Fig. 9. During summer months, higher PV availability results in SoC levels that remain closer to the upper bound, often achieving full charge in midday hours. In contrast, winter months present a more frequent cycling pattern with the deeper discharges, highlighting the increased reliance on storage to compensate the reduced PV output. The figure confirms also the appropriate size of the storage, as SoC shows both full-charge and near-minimum events, indicating a balanced use of the available capacity.

Table 5
BSS characteristics.

Quantity	Value
Useable nominal capacity	64 MWh
Autonomy	29 h
Battery wear cost	0.049 \$/kWh
Annual throughput	1.105 GWh/year

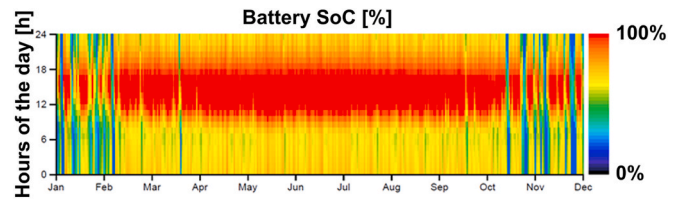


Fig. 9. SoC management during the year.

Operationally, this highlights the ability of the storage to buffer seasonal and daily variability, ensuring reliability of charging process, avoiding unnecessary oversizing. Finally, it is possible to structure a cost analysis of the system. In the methodology used it is possible to carry out the calculation in terms of Net Present Cost (NPC). In doing so, the costs of each of the components, the operating costs and the costs related to fuel and replacement are considered. Finally, the salvages that will occur over the investment period must be deducted from this investment (1,724,666 \$), which in the case study is considered 25 years, in compliance with the useful life of the PV panels. The total investment estimated through the simulation will be: G\$127,392. The investments incurred in terms of NPC are shown in Fig. 10.

Then, CoE is evaluated on the investment incurred, which takes into account the accumulation system and the project years, within which it is assumed that the investment spent will be recovered. Therefore it is reasonable that the energy cost of these systems is initially high. Following this logic, the simulation highlights as CoE produced by the system $\text{CoE} = 0.418 \text{ \$}/\text{kWh}$. However, it should be underlined that the authorities could provide incentives for decarbonization, reducing the cost of the investment and therefore also the levelized CoE. Similarly, in the identified case study there is a large surplus of energy production, which can be resold, generating extra profits which could benefit the economic balance sheet. Finally, the presence of an important storage system could be useful for the network manager who could pay for ancillary network services, which the PTO could favour thanks to the wide availability of storage systems providing flexibility.

After evaluating the two depots as aggregates, it is appropriate to return to the real case, dividing the power generated with that which would be used in the depots. Considering an occupied space of $6 \text{ m}^2/\text{kW}$ the equivalent PV plant will be approximately 0.126 km^2 . Knowing that the size of the depot in the Al Ma'mura is approximately 1.09 km^2 , and the depot located in the Al Qabari is approximately 0.8 km^2 , although not completely coverable with PV plates, both would be able to accommodate the PV system in its entirety. However, it is reasonable to divide production based on available space. Therefore, knowing that the total power of the depots (P_{tot}) will be 21.017 MW, the power is divided for each i^{th} depot ($P_{\text{depot},i}$) depending on the surface area ($A_{\text{depot},i}$) using (5). The power produced in Al Ma'mura depot is 12.143 MW, while the power produced by Al Qabari depot is 8.873 MW. Considering the estimated annual production from the microgrid of the two depots, through (9) 17,544 tons of avoided CO_2 emissions per year are expected

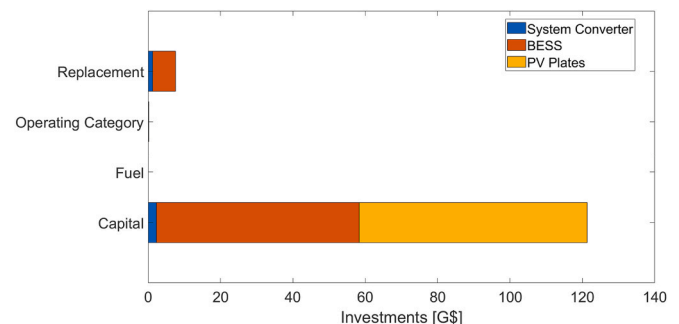


Fig. 10. NPC of the PV plant divided by cost type.

from the simulation. Then, considering (10) it is possible to highlight their impact in terms of CSV, obtaining 1.4 M\$/year. This helps to quantitatively highlight the social and economic impacts that plants of this size would have. In addition, the emission costs saved could go a long way toward offsetting the previously calculated cost of energy. All of these aspects combined with the shift to electric in the transportation section can yield significant benefits for citizens and residents. However, beyond these baseline results, it is essential to assess how the system behaves under varying conditions. For this reason, a discussion will be implemented supported by sensitivity analyses on demand growth and load variability, in order to test the robustness and scalability of the proposed solution.

5. Discussion

This section allows to expand the analysis beyond the baseline simulation to test the robustness and broader implications of the proposed methodology. While the results presented in Section 4 validated the technical feasibility and the economic sustainability of a PV-powered public transport system in Alexandria, it is necessary to investigate how these outcomes evolve under varying boundary conditions. This discussion introduces a multi-scenario sensitivity analysis, focusing on demand growth and load variability, and evaluates their impact on PV sizing, storage requirements, costs, and environmental indicators. By doing so, this section highlights the resilience of the system against uncertainties and identifies trade-offs between PV expansion and storage sizing. Finally, the inclusion of environmental and social indicators, such as avoided CO₂ emissions and CSV, enables a more comprehensive assessment of system sustainability.

Therefore, at first it is important to consider how the system will respond to growing energy demand. A multi-scenario approach was used to address this challenge. Three growth demand rates are evaluated.

- 1) Soft growth demand rate (+1 %);
- 2) Moderate growth demand rate (+3 %);
- 3) Aggressive growth demand rate (+5 %).

First, Fig. 11 is used to show how monthly power demand is positioned for each scenario compared to the production analyzed so far. The figure highlights the results of this growth in demand compared to the production assumed in the baseline scenario implemented in the case study.

The figure compares simulated monthly photovoltaic production and constant power requirements in different demand growth scenarios. The blue curve shows the typical trend in PV generation, with lower values in winter (3.2–3.4 MW) and a summer peak of between 5 and 5.2 MW, as already seen in Fig. 7. The constant horizontal lines represent the power requirements in the three growth demand rate scenarios, compared with the baseline scenario. The values do not exceed 1.7 MW per month,

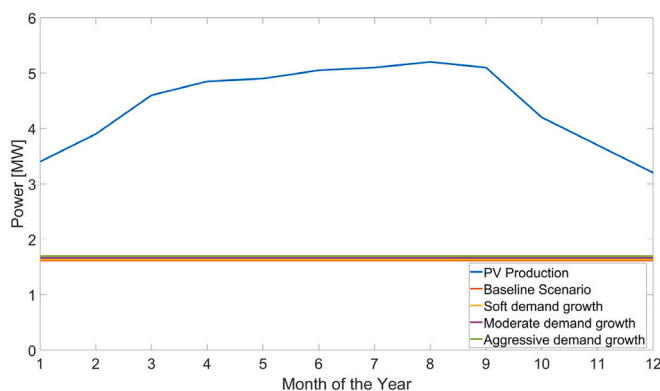


Fig. 11. Monthly PV power output produced in relation with power demand.

remaining below the production capacity of the photovoltaic system in all months of the year. This analysis allows us to confirm that solar production is seasonally variable but always exceeds the assumed demand. At the same time, it confirms that different demand growth scenarios do not substantially change the relationship between supply and demand, suggesting that the PV system remains capable of fully covering loads even in the presence of aggressive demand growth. By simulating this growth progressively, Table 6 helps us identify the changes that the system suggests to the plant, keeping load variability fixed at 10 %.

The results show that the increase in loads mainly affects the size of the PV source and, to a lesser extent, the storage system. In particular, installed PV capacity grows progressively from 21 MW to 21.4 MW in the soft scenario (+0.4 MW), 22.9 MW in the moderate scenario (+1.9 MW), and almost 30 MW in the aggressive scenario (+8.9 MW). At the same time, the number of batteries remains unchanged in the slow growth scenario, decreases slightly in the moderate scenario (77, −3 units compared to the baseline) and decreases markedly in the most aggressive scenario (59, −21 units compared to the baseline scenario). This behavior suggests that, as installed solar generation grows, optimization identifies a lower relative need for storage. This means that the availability of greater peak power and distributed generation during the day reduces the need to store energy to cover periods of deficit. From a system perspective, this indicates that investment in photovoltaic capacity not only responds to increased demand but also partially replaces storage. Therefore, in scenarios of more aggressive demand growth, the additional energy required can be directly met by expanding PV generation, reducing dependence on storage systems and mitigating the costs associated with their sizing and replacement. This trade-off between PV and storage is particularly evident in the aggressive demand growth scenario, where nearly 9 MW of additional PV reduces the number of storage units by more than 25 % compared to the baseline scenario. From an economic point of view, the NPC increases from 127.4 M\$ (baseline) to 139.1 M\$ in the aggressive scenario (+11.7 M\$), following the expansion of PV infrastructure. The LCOE, on the other hand, remains stable at 0.42–0.43 \$/kWh, confirming that the higher investment is offset by the greater amount of energy produced and consumed. Daily demand grows from 53.8 MWh in the baseline scenario to 56.5 MW in the most aggressive scenario, while peak power demand increases from 6.7 to 7.0 MW. The load factor remains constant, indicating that the regularity of system use is not compromised. This analysis shows that the growth in energy demand could already be offset by the existing system. On the other hand, optimizing the system based on demand growth would require an increase in PV generation, which would also reduce stress on stationary battery performance, as the reduction in storage requirements would help reduce degradation. At the same time, the economic impact is measurable in terms of NPC, which would grow by no more than 9 %, without however affecting the cost of the energy produced. These results suggest that the system maintains economic and energy resilience even in scenarios of growing demand. In addition to technical and economic indicators, the sensitivity analysis includes environmental and social aspects such as the avoided CO₂ emissions and the CSV. The avoided CO₂ emissions grow from 17.5 ktons/year in the baseline scenario to 18.4 ktons/year in the aggressive growth demand scenario (+0.87 avoided CO₂ ktons/year), showing how larger PV share consistently strengthens decarbonization outcomes. In parallel, the associated CSV increases from 1,40 M\$/year to 1.47 M\$/year, reflecting the monetized benefit of reduced emissions. These results confirm that demand growth scenarios, while requiring higher PV capacity and slightly higher investments, also yield proportional improvements in avoided emissions and social value, further supporting the sustainability of the proposed transition.

Finally, it should be reminded that the proposed system has intrinsic limitations. The electrical load is modeled on a LPT system which could be more variable than assumed, not using a single clocking and a single commercial speed, but varying during the day in a more unpredictable manner, also depending on the traffic in the city. This would increase the

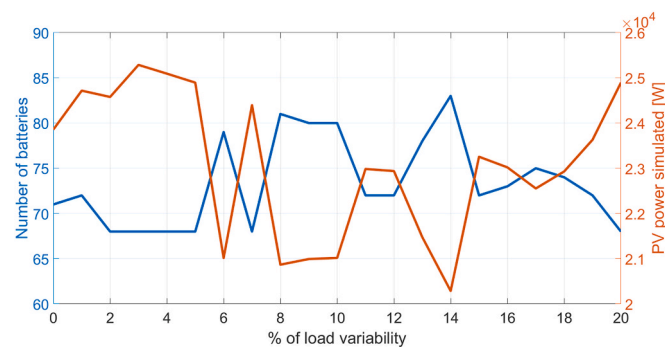
Table 6

System sensitivity analysis based on demand growth, with optimized performance.

Quantity	Baseline	Soft growth	Baseline – Soft growth variation	Moderate growth	Baseline – Moderate growth variation	Aggressive growth	Baseline – Aggressive growth variation
% demand increment	–	+1 %	+1 %	+3 %	+3 %	+5 %	+5 %
PV size [MW]	21.02	21.41	+0.36	22.89	+1.87	29.93	+8.91
Battery number	80	80	–	77	–3	59	–21
NPC [M\$]	127.39	128.70	+1.31	131.20	+3.81	139.08	+11.68
Levelized CoE [\$/kWh]	0.42	0.42	–	0.42	–	0.43	+0.01
Daily energy demand [MWh/day]	53.81	54.35	+0.54	55.42	+1.61	56.50	+2.69
Peak power [MWp]	6.67	6.74	+0.07	6.87	+0.2	7.01	+0.33
Load factor	0.34	0.34	–	0.34	–	0.34	–
Avoided CO ₂ emissions [ktonsCO ₂ /year]	17.54	17.72	+0.17	18.07	+0.52	18.42	+0.87
CSV [M\$/year]	1.40	1.42	+0.02	1.44	+0.04	1.47	+0.07

uncertainty of consumption of the TPL service, causing the sizing of the PV plant of the depots to vary. Precisely for this reason PV plant positioned on the depots is sized starting from a load that is affected by a daily variability of 10 %, so as not to incur the risk of unused energy. It is interesting to see how the load variability parameter impacts the system simulation. Specifically, how the power of the system and the number of batteries required vary to guarantee the service even on an island. Fig. 12 shows the trends of the simulated PV power and that of the number of batteries as the daily load variability increases. It is interesting to note how the two values, as expected from the analysis of Figs. 8 and 9, tend to compensate.

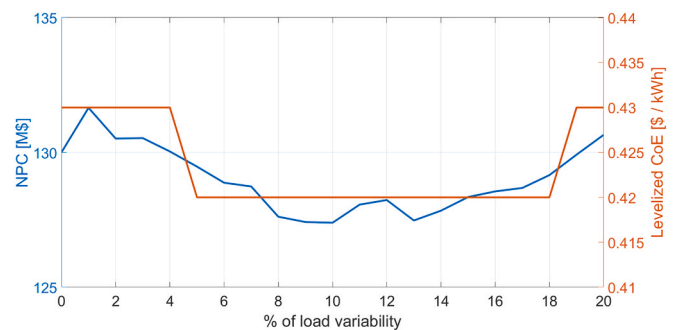
The result obtained in this scenario is not surprising. The trend of the number of batteries perfectly matches that of the simulated PV power. This behaviour is closely related to the CC, which optimizes the system size based on the power demand. In this regard, the system is optimally sized to supply the load, whether it is provided by PV panels or batteries, in accordance with (3). Consequently, for certain percentages of load variability, a larger storage bank may be required to compensate for the variability. However, it is not correct to state that increasing the variability should increase the number of batteries in the storage bank. In some load variability conditions, the optimal configuration to power the load and ensure that the batteries have a sufficient SoC requires expanding the PV plant, resulting in an increase in generated power. As in other cases, this power can either supply the load, be sold, or stored in the batteries. This process demonstrates that in addition to the storage bank configuration, the amount of annual throughput in the storage bank and the average SoC level throughout the year will also change with load variability. This process allows for the identification of the optimal system configuration. In Fig. 12, while the number of batteries fluctuates between 68 and 85 units without a clear monotonic trend, the PV capacity remain relatively stable in the range between 20 and 25.5 MW. This highlights that, even under higher demand uncertainty, the PV sizing is robust, whereas storage requirements are more sensitive and irregular, reflecting the balancing role of battery system under variable

**Fig. 12.** Impact of daily load variability in simulation in size of BSS (blue) and PV power output (orange).

load profile. Clearly, load variability has an impact also on other aspects of the case study. As a limitation, it must be noted that battery degradation is not explicitly modeled, as the round-trip efficiency and SOC window is kept constant without calendar or cycle aging. Therefore, storage sizing and CoE should be interpreted as lower-bound estimates subject to lifetime effects. Likewise, PV positioning is approximated, as PV is sized through the surface per kW assumption and depot-level checks without adopting detailed optimization, since the wide space availability. Fig. 13 highlights the impact on NPC required for the depot and levelized CoE of the electricity production.

Through Fig. 13 it is possible to connect load variability to the economic performances of the simulated system. The NPC varies between 127 M\$ and 132 M\$. The NPC peak is reached at 131.65 M\$, when daily load variability is 1 %. This happens when PV power production and batteries number have a bad trade-off fulfilling the power demand. On the other hand, the NPC is minimum when the load variability stays at 10 %, meaning the optimal compromise in terms of economics. This leads to a reduced levelized CoE produced by the simulated depot. In any case, CoE remains stable in the narrow band between 0.42 and 0.43 \$/kWh. This stability suggests that, although battery sizing adjustments influence the total investment cost, the effect on the cost of energy is marginal, supporting the resilience of the system economics under uncertainty in load profile. Finally, also electrical parameters such as peak load of the system and load factor must be investigated. Fig. 14 proposes their trend as a function of load variability.

Fig. 14 further explains the technical drivers of these trends. As load variability increases, the peak load rises from 3.9 MW up to 9.5 MW, while the load factor decreases from 0.57 to 0.23. The load factor deterioration means the system must handle sharper peaks with less uniform utilization, which helps to explain the oscillations observed in battery and PV size (Fig. 12) and the variations observed in NPC trends (Fig. 13). The overall evaluation of Figs. 12–14, shows how increasing the load variability simultaneously stresses the system from different perspectives. On the technical perspective, battery sizing becomes less

**Fig. 13.** Impact of daily load variability in simulation in NPC (blue) and Levelized CoE (orange).

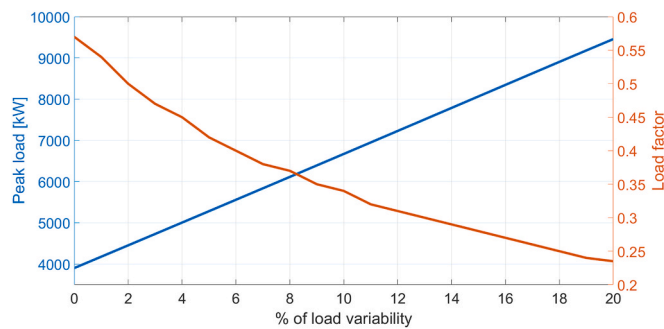


Fig. 14. Impact of daily load variability in simulation in peak load (blue) and load factor (orange) of the simulated system.

stable while PV power remains almost unaffected. On the economic perspective, this translates into moderate variations in NPC but nearly constant levelized CoE, highlighting that system economics are robust even under uncertain demand. Moreover, the peak load and load factor trends highlight that higher variability generates sharper peaks and lower utilization, forcing the storage system to absorb more irregular fluctuations. Alternatively, it could bring PV system to be oversized. Overall, the combined analysis demonstrates that while technical sizing is sensitive to load variability, the economic outcomes and energy cost remain resilient, thereby strengthening the confidence in the optimization results.

This discussion also clearly highlights the paper's contributions compared to prior work. It introduces both a sizing and operating methodology and a large-scale depot case study in a Mediterranean setting. The focus is on achieving self-sufficiency and nighttime coverage, very different from the grid-connected, cost-minimization approaches that dominate the existing literature. This dual contribution enables a comparative assessment with respect to studies cited in Table 1, while explaining why these results stay at the upper end for on-site PV utilization and CO₂ savings. The depot-scale design adopted in this work is larger and more effective as it is self-sufficiency-oriented than prior depot works that remain grid-tied and prioritize cost and emissions without schedule-aware night coverage. Studies [40,41], respectively on Fez and Malaysian depots demonstrate the economic and environmental benefits of PV-assisted charging under grid connection, whereas the proposed study emphasizes islanding capability and night operation under Mediterranean irradiance. City level LPT analyses [20] and PV-microgrid e-mobility in Sub-Saharan Africa [18,19,39] point in the same direction, cost and CO₂ reduction, but at different scales and objectives. The proposed schedule aware design explains the gap in PV size, storage and saved emissions. Overall, differences stem from resources, related to the Mediterranean solar radiation, the self sufficiency and grid-tied cost objective and the operational coupling between timetables and depots scheduling. Moreover, beyond baseline results, a two-dimensional sensitivity analysis that couples demand growth and daily load variability is performed, improving the robustness of the performed simulation. Considering the overall methodology, this schedule-aware, self-sufficiency-oriented sensitivity extends the typical grid-tied and focus depot studies by explicitly linking timetables to sizing, costs, and CO₂ savings outcomes.

6. Conclusion

This study presented a practical and scalable methodology for the

refurbishment of LPT systems through the integration of renewable energy sources and full electrification. By combining BRT and LRV services with optimally sized PV-based microgrids, the proposed approach enables a 100 % renewable-powered transport system. The case study conducted in Alexandria, Egypt, demonstrated the feasibility of electrifying two depots with PV systems of 12.143 MW and 8.873 MW, capable of not only meeting service demand but also generating profitable excess electricity. The system is projected to save approximately 17,544 tons of CO₂ annually, corresponding to \$1.4 million in avoided emission costs, thereby aligning with global net-zero emission objectives. Sensitivity analysis confirms robustness across demand growth scenarios (+1 % to +5 %), maintaining stable COE (0.42–0.43 \$/kWh) and increasing CO₂ savings to 18.42 ktons/year.

The proposed methodology offers notable advantages, including its simplicity of implementation, scalability to different transport networks, and compliance with current decarbonization policies. Importantly, it achieves these benefits without requiring extensive passenger behavior data, which facilitates rapid deployment. However, this simplification also represents a limitation, as the absence of demand-based passenger data may reduce timetable precision and lead to variability in operational efficiency.

Future work should focus on integrating passenger demand modeling and exploring additional uses for excess electricity, such as feeding local grids or supporting ancillary services, to further enhance economic sustainability and accelerate the return on investment. With these refinements, the methodology can serve as a replicable model for sustainable, user-oriented urban mobility, contributing significantly to the transition toward carbon-neutral cities.

CRediT authorship contribution statement

Cristian Giovanni Colombo: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Seyed Mahdi Miraftebadeh:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Michela Longo:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A



Fig. A1. a) Renewed BRT system, b) Renewed LRV system.

Table A1

Monthly km served, monthly required energy and minimum number of vehicles for BRT service

	Speed [km/h]	15			20			25		
	Clocking [min]	5	10	15	5	10	15	5	10	15
Monthly km Served	BRT 1	472563	249216	158592	470268	234864	157644	481884	237888	162210
	BRT 2	285792	141120	91680	279600	139800	91584	274752	137376	92088
	BRT 3	461916	241956	153972	454752	227376	152244	468546	230958	157620
Monthly Energy Required [kWh]	BRT 1	826985	436128	277536	822969	411012	275877	843297	416304	283868
	BRT 2	500136	246960	160440	489300	244650	160272	480816	240408	161154
	BRT 3	808353	423423	269451	795816	397908	266427	819956	404177	275835
Min Buses/day	BRT 1	84	44	28	96	48	36	78	42	30
	BRT 2	52	28	20	60	30	24	48	24	18
	BRT 3	84	44	28	96	48	36	78	42	30

Table A2

Monthly km served, monthly required energy and minimum number of vehicles for LRV service

	Speed [km/h]	15		20		25	
	Clocking [min]	10	20	10	20	10	20
Monthly km Served	LRV 1	92610	45360	91350	45360	90720	45360
	LRV 2	89788	43978	88588	43978	87937	43978
	LRV 3	88200	43200	87000	43200	86400	43200
	LRV 4	98814	48384	97440	48384	96768	48384
Monthly Energy Required [kWh]	LRV 1	143546	70308	141593	70308	140616	70308
	LRV 2	139171	68165	137311	68165	136303	68165
	LRV 3	136710	66960	134850	66960	133920	66960
	LRV 4	153162	74995	151032	74995	149990	74995
Min Trams/day	LRV 1	14	8	10	6	8	4
	LRV 2	14	8	10	6	8	4
	LRV 3	14	8	10	6	8	4
	LRV 4	14	8	10	6	8	4

Data availability

The authors do not have permission to share data.

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